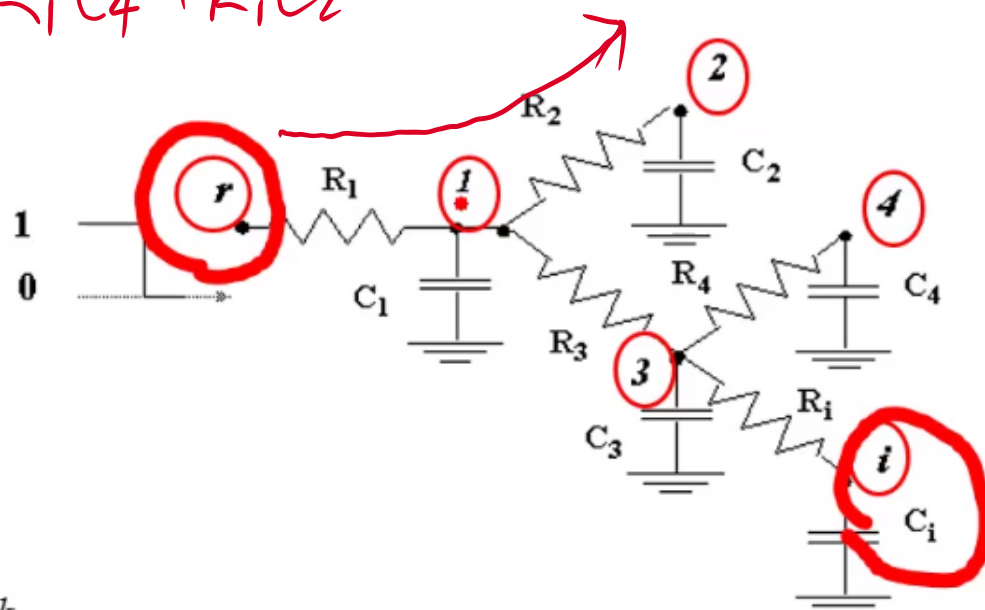


① $r \rightarrow 2$

集总RC模型

$$\tau_{r2} = R_1 C_1 + (R_1 + R_2) C_2 + R_1 C_3 + R_1 C_4 + R_1 C_i$$

$$t_{r2} = 0.69 \tau_{r2}$$



$$\tau_{D2} = \sum_{K=1}^N C_k R_{ik}$$

$$= R_1 C_1 + R_1 C_2 + (R_1 + R_3) C_3 + (R_1 + R_3) C_4 + (R_1 + R_3 + R_i) C_i$$

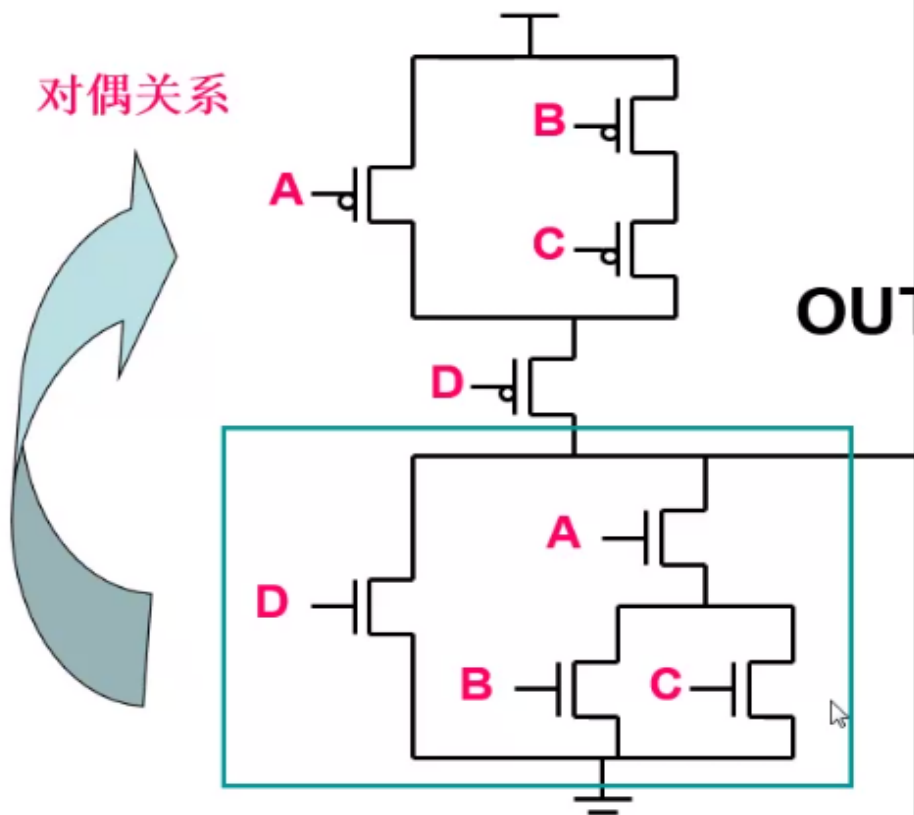
$$t = 0.69 \tau_{AB}$$

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简答：反相器的特点：

- 输出高电平和低电平分别为 V_{DD} 和 GND （噪声容限↑）
- 逻辑电平与器件的相对尺寸无关（无比逻辑）
- 稳态时在输出和 V_{DD} 或 GND 之间总存在一条具有有限电阻的通路，即具有低的输出阻抗。（抗噪声和干扰↑）
- 输入电阻极高，扇出系数很大（压控器件）
- 稳态下 V_{DD} 和 GND 之间没有直接的电流通路，即没有任何静态功耗。

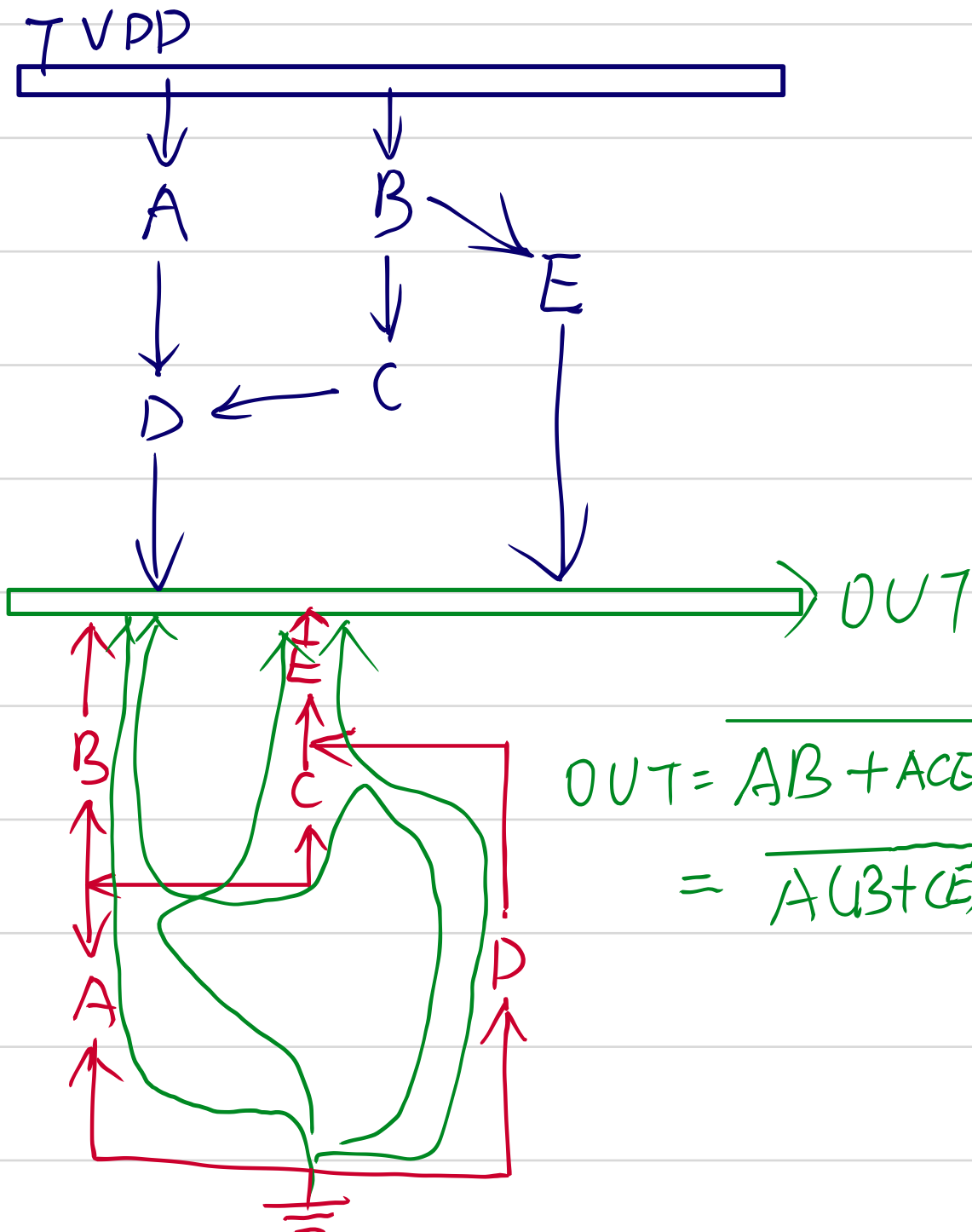
Complex CMO



$$D \cdot (A + BC)$$

$$D + A \cdot (B + C) \quad \checkmark$$

$$OUT = \overline{D + AB + AC} \quad \checkmark$$



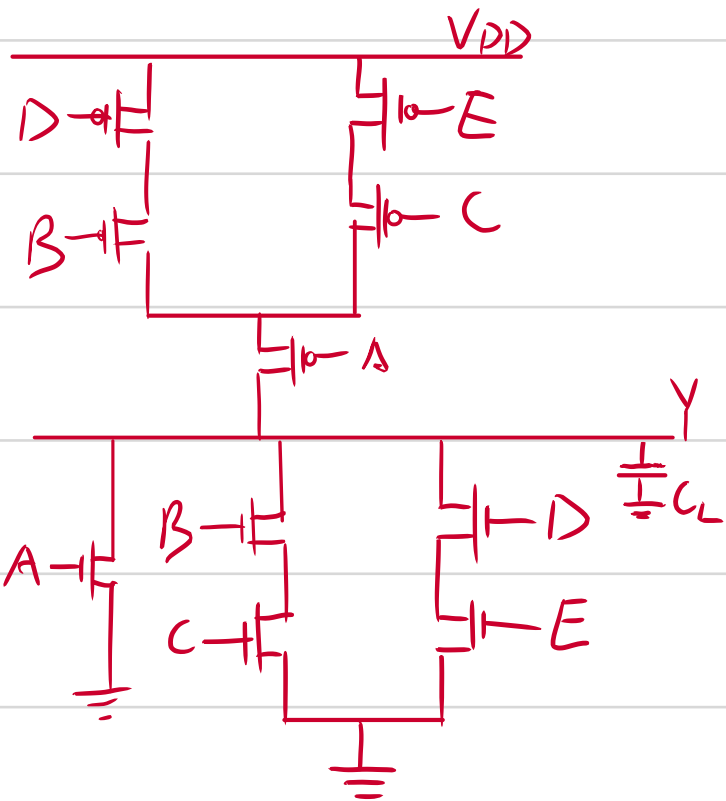
$$OUT = \overline{AB + ACE + DE + DCB}$$

$$= \overline{A(B + CE) + D(B + E)}$$

设计题(本题共 6 题，每小题 10，满分 60 分)

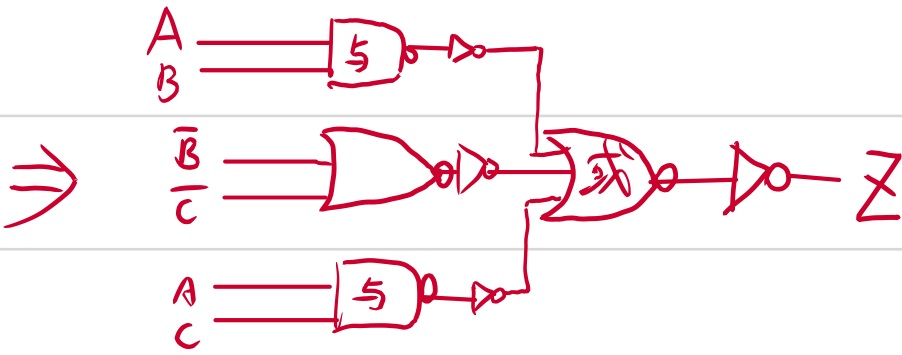
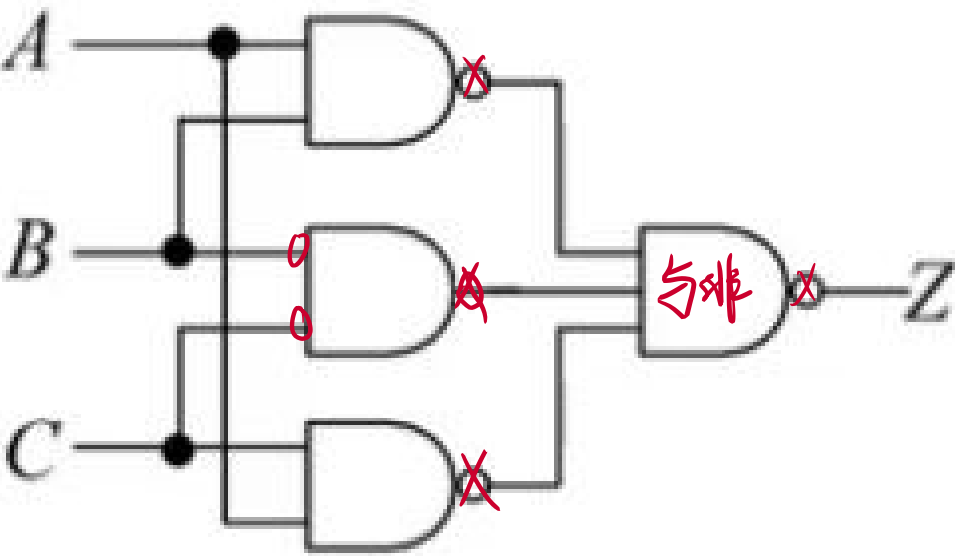
1. 利用互补 CMOS 结构设计有下列逻辑关系的电路图。 根据 PMOS 管的导通电阻为 R_P ，NMOS 管的导通电阻为 R_N ，负载电容为 C_L ，计算平均边沿延迟时间 $t_p=0.5(t_{pLH}+t_{pHL})$

$$Y = A + BC + DE$$



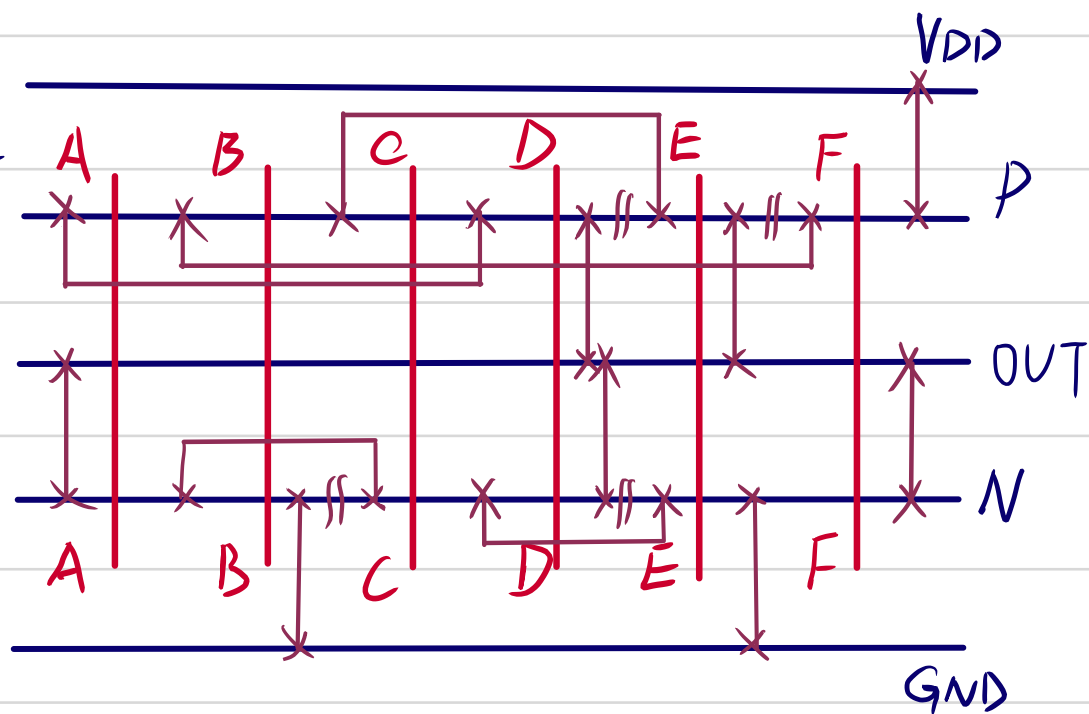
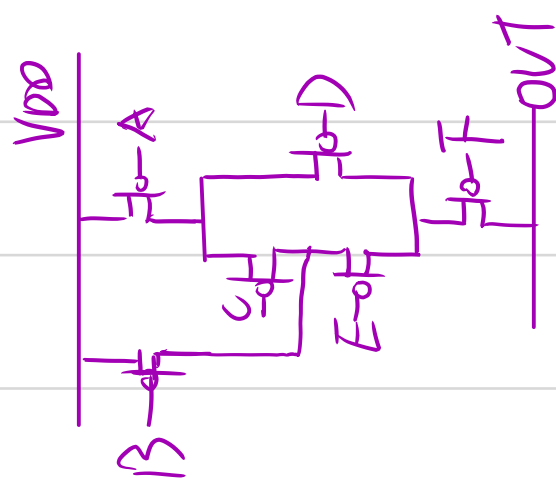
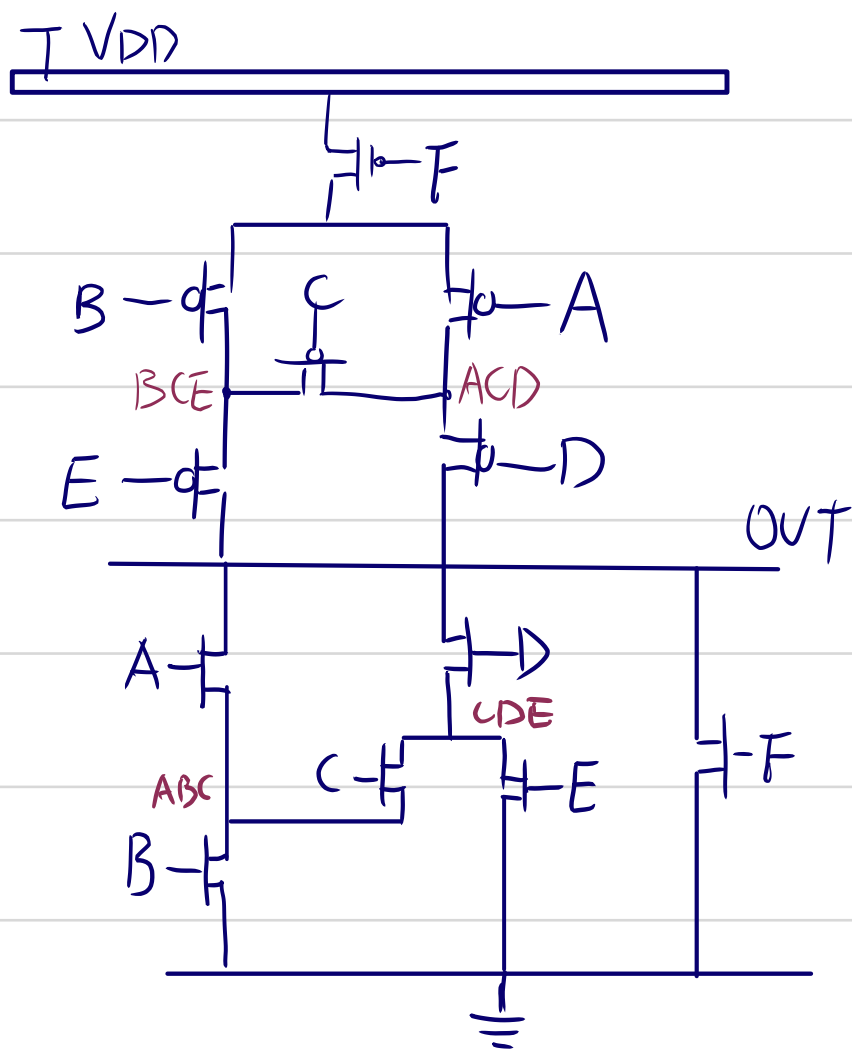
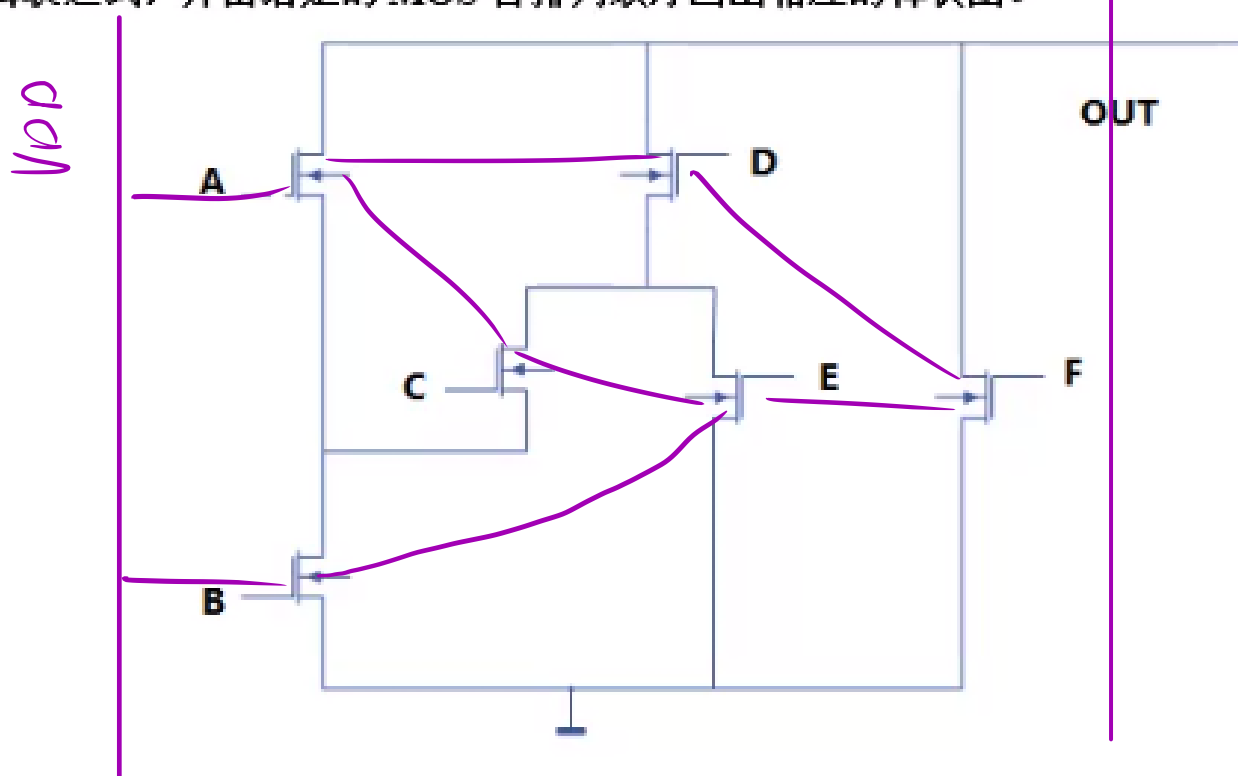
$$t_{pLH} = 0.69 R_P \times C_L \times 3$$
$$t_{pHL} = 0.69 R_N \times C_L \times 2$$
$$t_p = 0.5 (t_{pLH} + t_{pHL})$$

3. 已知电路如下图所示，试用多米诺逻辑电路实现图示电路的功能。



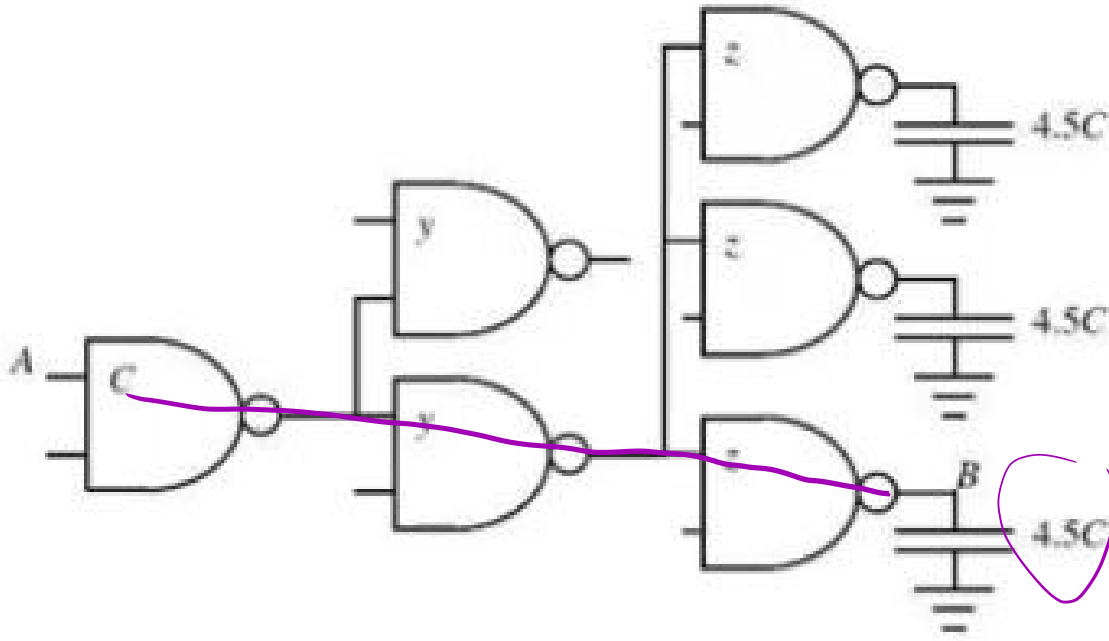
$$Z = AB + (\overline{A} + \overline{B})C$$

4. 根据图示 NMOS 网络电路图，利用相关知识，补全画出相应的 PMOS 网络电路，写出 OUT 的逻辑表达式，并由给定的 MOS 管排列顺序画出相应的棒状图。



$$OUT = B(A+D) + E(AC+D) + F$$

2. 分别计算从 A 点到 B 点的逻辑努力 G (logic effort); 总的分支努力 B (total branching effort), 电气努力 F (electrical effort)。设反相器的输入电容为 $1C$, 其它逻辑电路的输入电容如图所示分别为 c, y, z 。



逻辑努力 $g_1 = g_2 = g_3 = \frac{4}{3}$ (查表)

NAND: $\frac{n+2}{3}$

电气努力 $F = \frac{C_L}{C_{g1}} = \frac{4.5C}{1C} = 4.5$

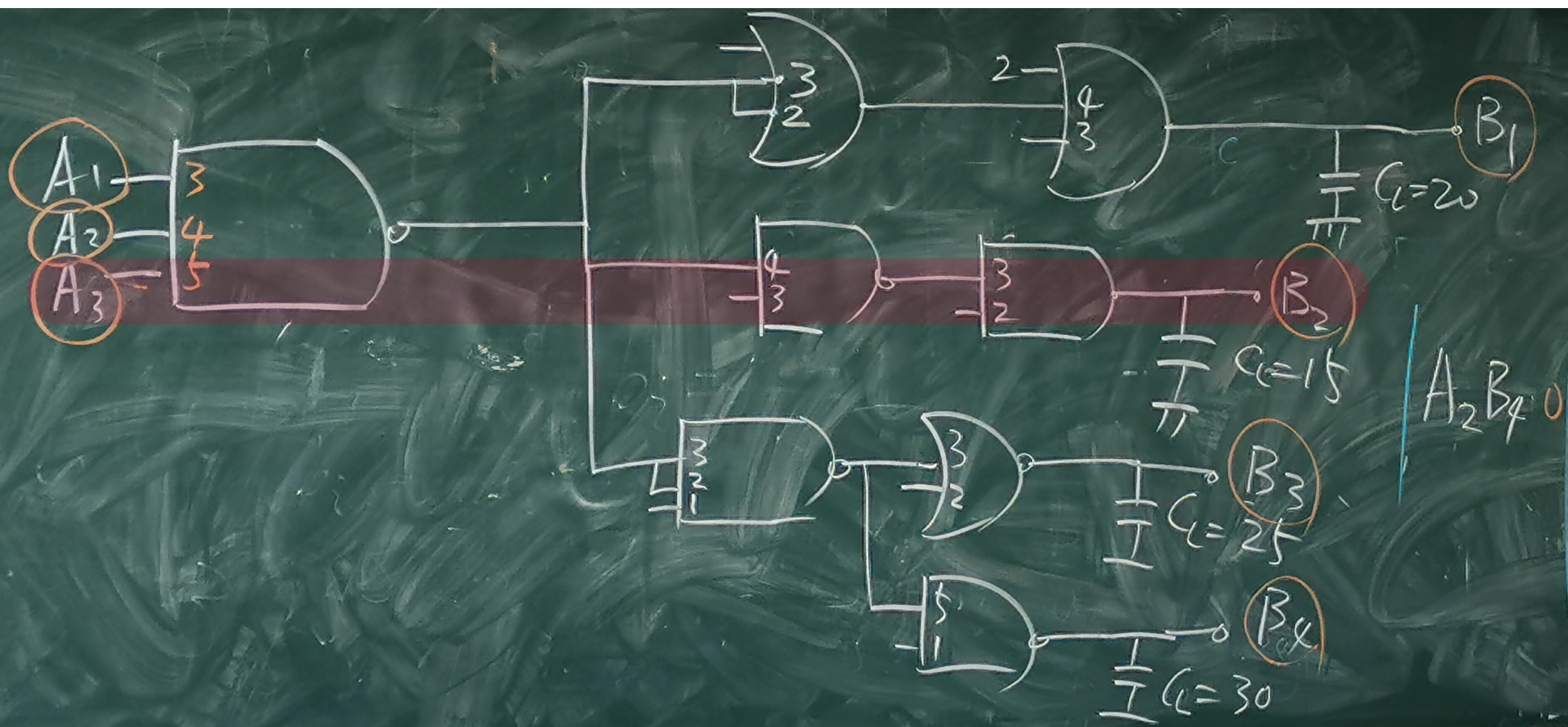
NOR: $\frac{2n+1}{3}$

$G = \prod_i g_i = 1 \times \frac{4}{3} \times \frac{4}{3} \times \frac{4}{3} = \frac{64}{27}$

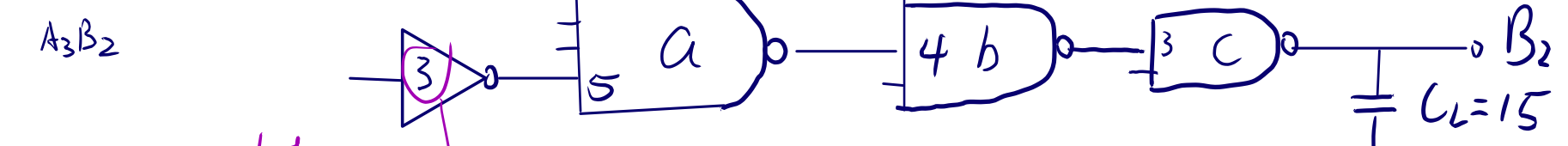
分支努力 $b_1 = \frac{2y}{y} = 2$ $b_2 = \frac{3z}{z} = 3$

$B = \prod_i b_i = 1 \times 1 \times 2 \times 3 = 6$

$H = F \times G$ $h = \sqrt[n]{H}$



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逻辑努力 g : $g_1 = \frac{3}{3} = 1$ $g_2 = \frac{5}{3}$ $g_3 = \frac{4}{3}$ $g_4 = \frac{3}{3} = 1$

电气努力 $F = \frac{C_L}{C_{g1}} = \frac{15}{3} = 5$

逻辑努力 $G = \prod_{i=1}^4 g_i = 1 \times \frac{5}{3} \times \frac{4}{3} \times 1 = \frac{20}{9}$

分支努力 $b_1 = \frac{C_{on-path} + C_{off-path}}{C_{on-path}} = \frac{(3+2) + 4 + (3+2)}{4} = \frac{14}{4} = \frac{7}{2}$ (同级的比)

$B = \prod_{i=1}^4 b_i = \frac{14}{4} \times 1 \times 1 \times 1 = \frac{14}{4} = \frac{7}{2}$

总路径努力 $H = GBF = \frac{14}{9}$

$h = \sqrt[4]{H} = 1.12$

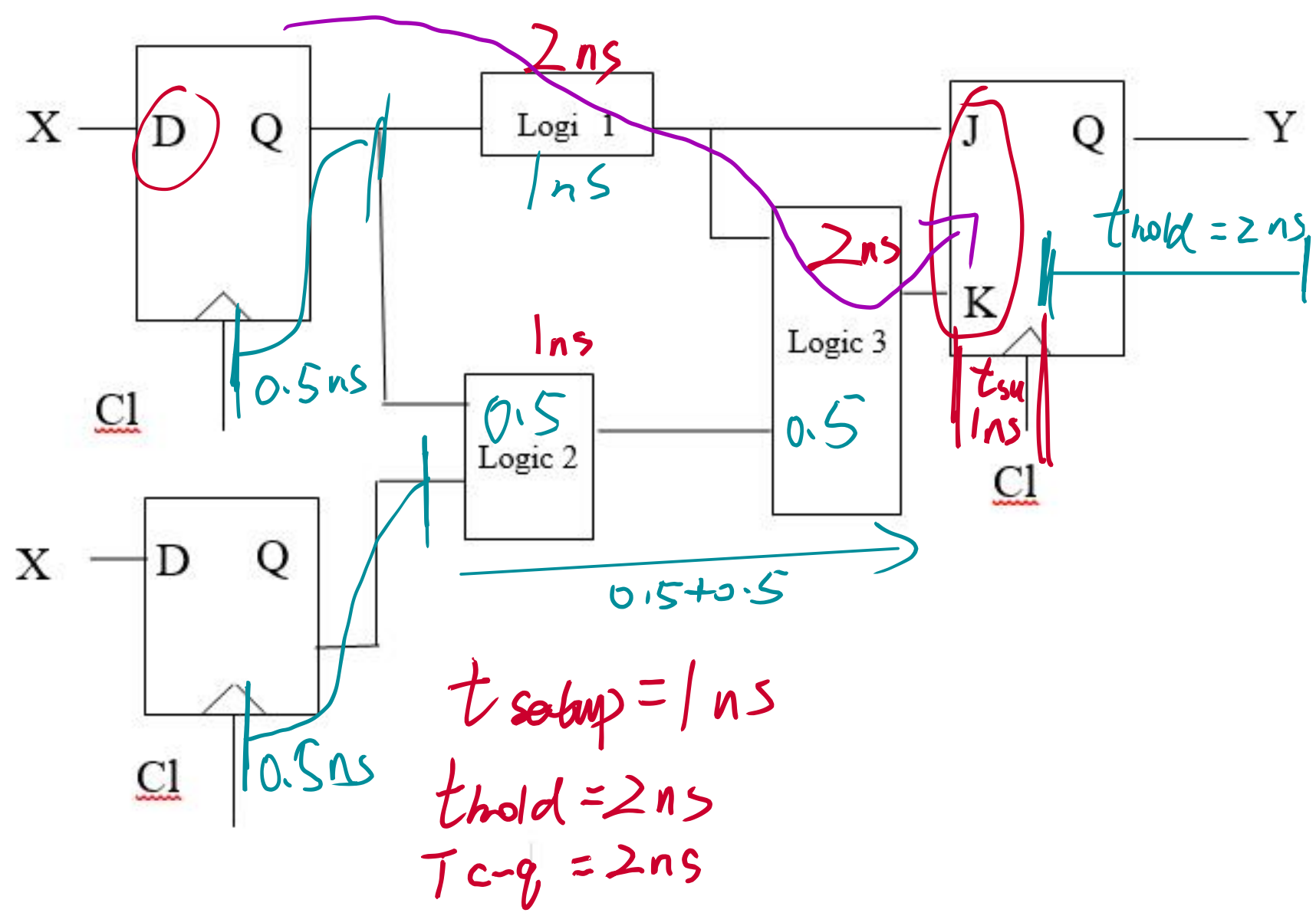
$\therefore f_1 = \frac{2.08}{1} = 2.08$ $f_2 = \frac{2.08}{\frac{5}{3}} = 1.248$ $f_3 = \frac{2.08}{\frac{4}{3}} = 1.56$ $f_4 = \frac{2.08}{1} = 2.08$

$f_1 = \frac{5}{3}$ $f_2 = \frac{5+4+5}{5} = \frac{14}{5}$ $f_3 = \frac{3}{4}$ $f_4 = \frac{15}{c}$ (前后级的比)

尺寸 $a = \frac{f_1 g_1}{g_2} = 1.248$ 尺寸 $b = \frac{f_1 f_2 g_1}{g_3} = 1.95$ 尺寸 $c = \frac{f_1 f_2 f_3 g_1}{g_4} = 4.05$

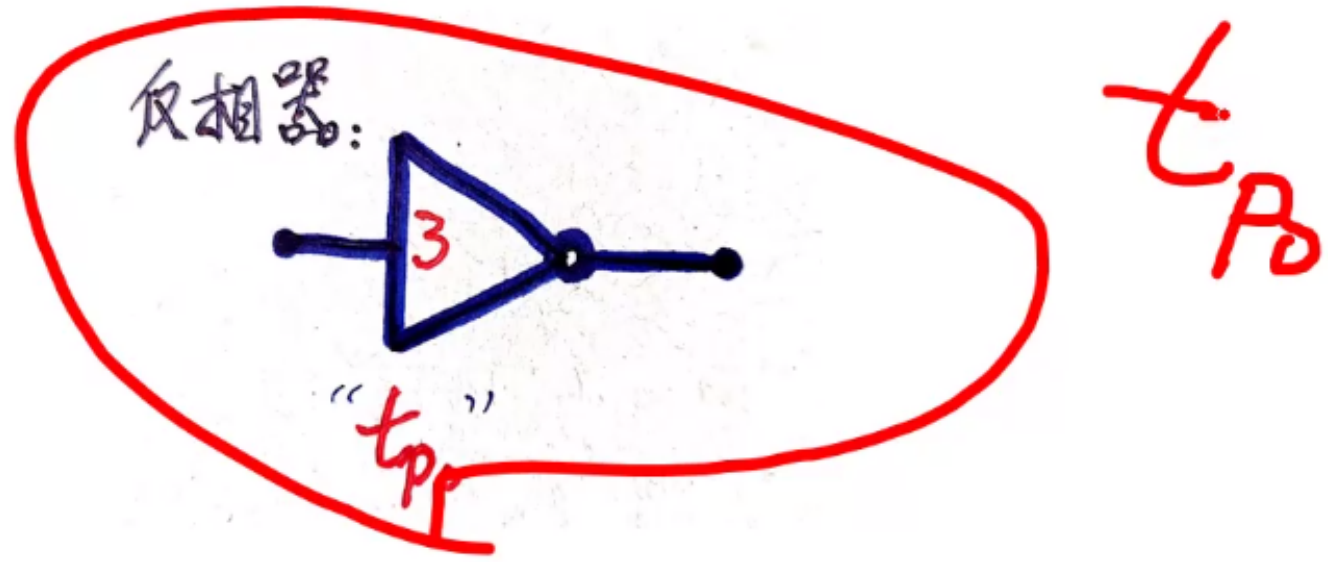
$s_i = \left(\frac{g_i s_i}{g_i} \right) \prod_{j=1}^{i-1} \left(\frac{f_j}{b_j} \right)$

如下图示，两触发器时间参数相同，其 T_{setup} 为 1ns， T_{hold} 为 2ns， T_{cq} 为 2ns，污染延时 T_{cd} 为 0.5ns；两个触发器之间为组合逻辑电路，由三部分组成：logic1、logic2 和 logic3，logic1 的最长延时为 2ns，污染延时为 1ns；logic2 的最长延时为 1ns，污染延时为 0.5ns；logic3 的最长延时为 2ns，污染延时为 0.5ns。请问该电路的周期为多少？是否存在时序违规问题，为什么？。



(1) $t_{c-q} + t_{su} + t_{plogic} \leq T$
 $2 + 1 + 4 = 7ns \leq T \therefore T = 7ns$ 找最慢的算周期

(2) $t_{cdreg} + t_{cdlogic} \geq t_{hold}$
 $0.5 + (0.5 + 0.5) \geq 2$
 $1.5 < 2$
 $\therefore$ 违规 找最快的算是否违规



反相器的本征延时 $\rightarrow$ 电气努力 $f = \frac{C_{ext}}{C_g}$

逻辑努力

$$t_p = t_{p0} \cdot \left(P + \frac{f \cdot g}{\gamma} \right)$$

与端口数量有关 查表 $\rightarrow \gamma \approx 1$

反相器:

$t_p = t_{p0} \cdot \left(P + \frac{f \cdot g}{\gamma} \right)$

逻辑努力 $\rightarrow \gamma \approx 1$

与端口数量有关 查表

$t_{p2} = t_{p0} \cdot \left(P + \frac{f \cdot g}{\gamma} \right) = 10 \cdot 3 = 30$

腾讯课堂

$z = R_1 C_1 + (R_1 + R_2) C_2 + (R_1 + R_2 + R_3) C_3 + R_1 (C_4 + C_5)$

$t_B = R_1 C_1 + (R_1 + R_2) C_2 + (R_1 + R_2 + R_3) C_3 + R_1 (C_4 + C_5)$

(集总RC型)

$t_p = 0.38 R \cdot C L^2$

$t_{p1} = t_{p0} \cdot \left(P + \frac{f \cdot g}{\gamma} \right) = t_{p0} \cdot \frac{16}{3}$

$t_{p2} = t_{p0} \cdot \left(P + \frac{f \cdot g}{\gamma} \right) = t_{p0} \cdot \frac{16}{3}$

$t_{p5} = t_{p0} \cdot \left(P + \frac{f \cdot g}{\gamma} \right) = t_{p0} \cdot \frac{20}{3}$

$t_{p3} = t_{p0} \cdot \left(P + \frac{f \cdot g}{\gamma} \right) = t_{p0} \cdot \left(4 + \frac{C_2}{3} \right)$

$t_{p4} = t_{p0} \cdot \left(P + \frac{f \cdot g}{\gamma} \right) = t_{p0} \cdot \left(3 + \frac{C_3}{3} \right)$

$t_{p6} = t_{p0} \cdot \left(P + \frac{f \cdot g}{\gamma} \right) = t_{p0} \cdot \left(2 + \frac{7}{3} \right) = t_{p0} \cdot \frac{13}{3}$

$t_{p7} = t_{p0} \cdot \left(P + \frac{f \cdot g}{\gamma} \right) = t_{p0} \cdot \left(3 + \frac{C_4}{3} \right)$

t_P

A.

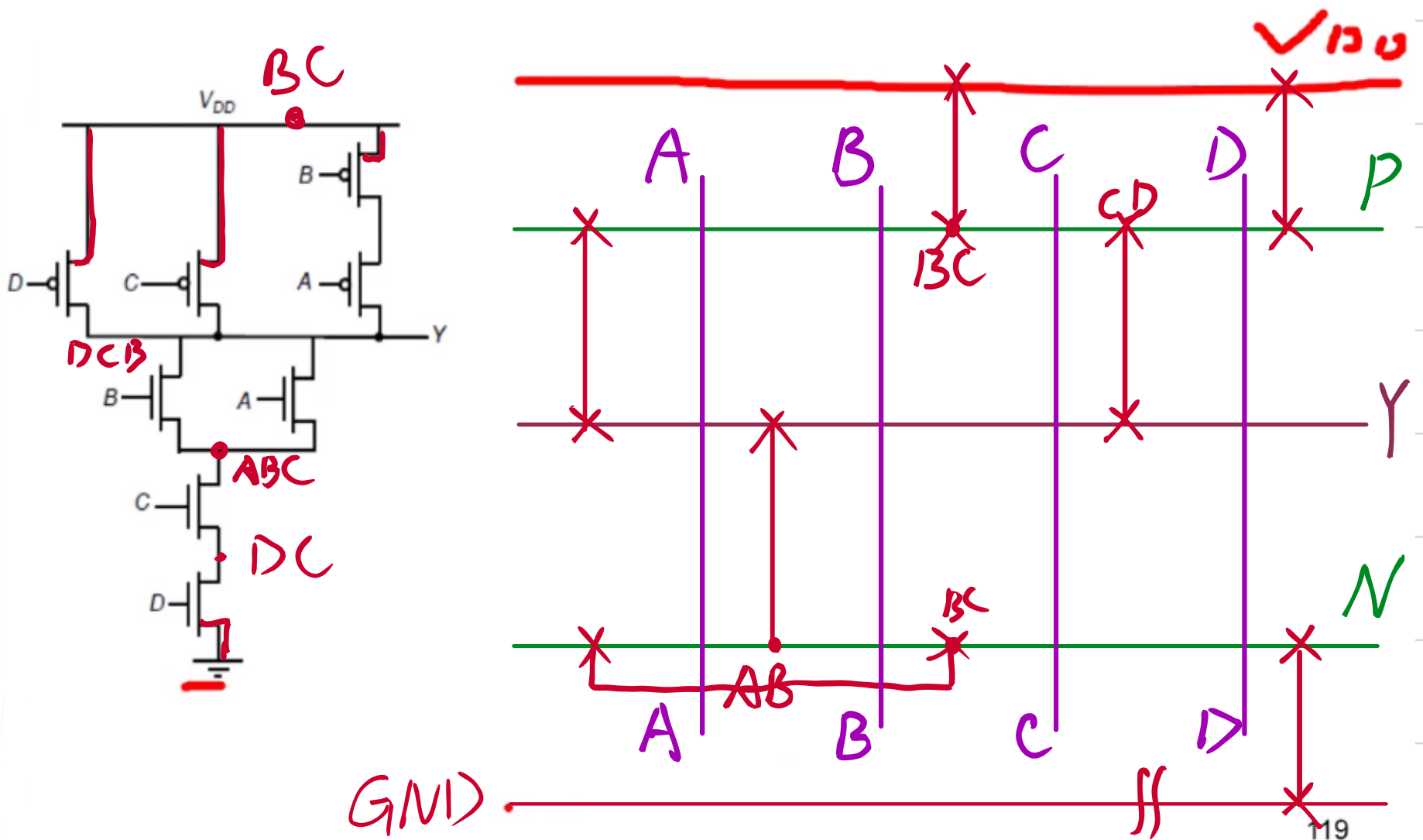
B

C

D

E

$$Y = \overline{CD(A + B)}$$



$$Y = \overline{CD(A + B)}$$

